(iii) Let $I = \int \operatorname{cosec} x dx = \dfrac{\int \operatorname{cosec} x(\operatorname{cosec} x - \cot x)}{\operatorname{cosec} x - \cot x} dx$

$$= \int \frac{\operatorname{cosec}^2 x - \operatorname{cosec} x \cot x}{\operatorname{cosec} x - \cot x} dx$$

Putting $\operatorname{cosec} x - \cot x = u$, then $(\operatorname{cosec}^2 x - \operatorname{cosec} x \cot x) dx = du$

$$\text{Thus, } I = \int \frac{1}{u} du = \log |u| + c = \log |\operatorname{cosec} x - \cot x| + c.$$

(iv) Let $I = \int \sec x dx = \int \dfrac{\sec x(\sec x + \tan x)}{\sec x + \tan x} dx = \int \dfrac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} dx$

Putting $\sec x + \tan x = u$, then $(\sec^2 x + \sec x \tan x) dx = du$

$$\text{Thus, } I = \int \frac{1}{u} du = \log |u| + c = \log |\sec x + \tan x| + c$$

Therefore, $\int \sec x dx = \log |\sec x + \tan x| + c.$

Thus the following are the important standard results.

(1)	$\int \tan x \, dx = \log \lvert \sec x \rvert + c$
(2)	$\int \cot x \, dx = \log \lvert \sin x \rvert + c$
(3)	$\int \operatorname{cosec} x \, dx = \log \lvert \operatorname{cosec} x - \cot x \rvert + c$
(4)	$\int \sec x \, dx = \log \lvert \sec x + \tan x \rvert + c$

Example 11.32

Integrate the following with respect to x.

(i) $\int \dfrac{2x+4}{x^2+4x+6} dx$ (ii) $\int \dfrac{e^x}{e^x - 1} dx$ (iii) $\int \dfrac{1}{x \log x} dx$

(iv) $\int \dfrac{\sin x + \cos x}{\sin x - \cos x} dx$ (v) $\int \dfrac{\cos 2x}{(\sin x + \cos x)^2} dx$

Solution

(i) Let $I = \int \dfrac{2x+4}{x^2+4x+6} dx$

Putting $x^2 + 4x + 6 = u$, then $(2x+4) dx = du$

$$\text{Thus, } I = \int \frac{du}{u} = \log |u| + c = \log |x^2 + 4x + 6| + c$$

$$\text{Therefore, } \int \frac{2x+4}{x^2+4x+6} dx = \log |x^2 + 4x + 6| + c.$$

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